MBA Fastrack 2025 (CAT + OMETs)

QUANTITATIVE APTITUDE

DPP: 9

Time Speed Distance III + Averages

Q1 In a race, A beats B by 100 m , B beats C by 200 m and A beats C by 290 m. In the same racing track P beats Q by 200 m, Q beats R by 150 m, then by what distance (in metres) P will beat R ?

Q2 Virat and Rohit begin walking together on a circular track from the same starting point simultaneously. If they walk in the same direction, Virat completes precisely 2 more laps than Rohit in 40 minutes. When they walk in opposite directions, they first meet after precisely 4 minutes. How many seconds it takes for Virat to complete one lap?

Q3 Trisha and Bhavna jog on a circular track of length 40m. They started from the same starting point at the same time and met for the first time after 20 seconds. Even if they jog for all day long, they will meet with each other at the same exact point. Bhavna's speed is 20% more than that of Trisha. Find the amount of time (in seconds) taken by Trisha to complete 1 round of the track

Q4 Timon and Pumba are located at the same point on a circular track of 6 kilometres. When they start running in the same direction, Pumba completes 5 more rounds than Timon in 50 minutes. If they run in opposite directions, Timon covers 1 kilometre less than Pumba on their first meeting. Find the speeds of Timon and Pumba respectively.
(A) 54 kmph, 90 kmph
(B) 90 kmph, 54 kmph
(C) 90 kmph, 126 kmph
(D) 144 kmph, 108 kmph

Q5 Two friends simultaneously start swimming towards each other from two edges exactly opposite each other of a pond. They meet at a distance of 320 m from one of the edges and continue swimming further till they reach the opposite edges respectively. They take rest for 1 minute each and reach back by taking the same path. Now they meet at a distance of 140 m from the other edge. Find the distance (in m) between the two edges of the pond.

Q6 Amit and Bhavna start jogging from the same point on a circular track at the same time. If they run in the same direction, they meet for the first time after 150 seconds. However, if they run in opposite directions, they meet for the first time after 25 seconds. After how much time will they both be back at the starting point together for the first time?
(A) 3 min
(B) 5 min
(C) 24 min
(D) Cannot be determined

Q7 P, Q, and R compete in a 1200-meter race. P can give Q a head start of 80 meters and R a head start of 200 meters. They now run a new race where Q starts 9 minutes after R, and P starts sometime later. All three finish the race exactly at the same time. How many seconds after R does P start running?

Q8 In a 1200-meter race, even if Arjun gives Rohan a 120-meter head start, Arjun still wins by 12 seconds. Additionally, if Arjun gives Rohan a 16-second head start, Arjun still finishes 60 meters ahead. Find Arjun's speed in meters per second.

Q9 X and Y start from diametrically opposite points on a circular track of 500 meters. X runs clockwise while Y runs counterclockwise at speeds of 15 m/s and 10 m/s, respectively. Determine how many times they will cross each other in the next 350 seconds.

Q10 In an 800-meter race, X beats Y by 40 meters and Z by 100 meters. They compete again in a 400-meter race, but this time, X starts 30 meters behind the starting line, and Z starts 20 meters ahead. Who wins the race?
(A) X
(B) Y
(C) Z
(D) Two of the persons will finish at the same time.

Q11 A student studied four subjects in his bachelor's degree. If the ratio of marks in subjects A and B is 1 : 2, in subjects B and C is 2 : 5 and in subjects A and D is 1 : 4. If she has scored an average of 84 marks in all the subjects combined. In how many subjects did she score greater than or equal to 60 marks?

Q12 Present ages of A and B are in the ratio 7: 5. Present ages of C and D are in the ratio 2: 3. Present average age of A, B and D is 36 years and the present average age of A, C and D is 34 years. Find the difference between the ages of A and D.
(A) 3 years (B) 6 years
(C) 9 years (D) 12 years

Q13 The average weight of three men A, B and C is 65 kg. Another man D joins the group and the average now becomes 68 kg. If another man E whose weight is 4 kg less than that of D, replaces A then the average weight of B, C, D and E becomes 67 kg. Find the weight of A (in kg).

Q14 Total number of students in three sections 'A', 'B' and ' C ' of class V is 120 . The average marks obtained by all the students in these three sections combined is 54.5 , Average marks obtained by students of section B and C taken together is $56\frac{2}{3}$ and the average marks obtained by students of section A is 48 . Find the number of students in section A

Q15 'A', 'B', 'C' and 'D' are four friends having distinct ages such that average age of three of them taken together is 43 years, 33 years, 38 years and 42 years. Find the difference between the age of the youngest and the oldest among the four friends.
(A) 28 years (B) 24 years
(C) 32 years (D) 30 years

Q16 There are 2 classes X and Y in a college. The average weight of the students in class X and class Y combined is 75 kg. The average weight of students in class X is 45 kg more than the average weight of students in class Y. Which of the following cannot be the ratio of the number of students in classes X and Y, if it is known that the average weights of students of both the classes are integers?
(A) 8 : 1 (B) 1 : 2
(C) 1 : 5 (D) 2 : 1

Q17 There are 50 students in a class having equal number of girls and boys. 10 new boys and 10 new girls, all of whom have scored equal number of marks are added to the class. The average marks of the boys in the class increases by 20% while the average marks of the girls decreases by 20%. What was the ratio of the average marks scored by the boys and girls respectively before the new students joined?
(A) 3 : 17 (B) 5 : 19
(C) 2 : 13 (D) 7 : 20

Q18 There are 10 students in the class. The average weight of 10 students decreases by 3 kg when the two heaviest students from the group are removed. If the average weight of the two heaviest students is $14\frac{6}{11}$% more than the average weight of the whole class, then the average weight of the remaining 8 students (in kg) is:

(A) 79.5 (B) 78.5
(C) 78 (D) 80.5

Q19 A certain number of vans were initially estimated to transport 120 boxes of books from a warehouse. However, it was later found that each van could only carry 2 boxes less than the initially estimated amount, and as a result, 5 additional vans were needed. How many vans were originally estimated to be required?

(A) 10 (B) 12
(C) 15 (D) 18

Q20 The average of 19 distinct natural numbers arranged in ascending order is 35. The average of the first 10 numbers is 21, and the average of the last 10 numbers is 48. What is the maximum possible value of the eighteenth number?

Q21 The average height (in inches) of n players of a basketball team is an integer. When 4 new players join the team, the average height of the team increases by 3 inches, and the total height increases by 25 feet. What is the highest possible average height of the team (in inches) after the 4 new players have joined?

Q22 Ravi, Mohan, and Suresh had taken a certain number of exams. The average marks obtained by Ravi, Mohan, and Suresh were 67, 81, and 70 respectively. The average marks obtained by Ravi and Mohan were 73, and the average marks obtained by Mohan and Suresh were 76. What is the approximate average of the marks obtained by all three of them taken together?

(A) 70.8 (B) 71.3
(C) 71.8 (D) 72.2

Q23 A company has three departments: X, Y, and Z. The average productivity score of Department Y is 18 points higher than that of Department Z. The number of employees in Departments X, Y, and Z are in the ratio 7:8:9, respectively. If the overall average productivity score of the company is 4 points less than the average score of Department X, find the approximate difference between the average scores (in points) of Departments X and Z.

(A) 12 (B) 13
(C) 14 (D) 15

Q24 10 years ago, the average age of a family of 5 members was 40 years. Two years later from then, two children were born. One year after that, a male member got married. The average age of the family after two years from then was 32 years. What is the present age (in years) of the wife of the newly married member ?

(A) 30 (B) 29
(C) 28 (D) 27

Q25 There are five stones and each of them has a distinct integral weight. If two stones are weighed at a time then the weights obtained in kilograms are 76, 77, 79, 80, 82 , 83, 84, 85, 86 and 88 then what is the absolute difference beteeen the weight of the heaviest stone and the lightest stone in kilograms?

Q26 The average of n natural numbers is 20. If x_1, the smallest number, is increased by 20, and x_n, the largest number, is decreased by 5, the new average becomes 25. Determine the value of n

Q27 In a team of 68 members, the average experience level is 188 months. The team consists only of developers and designers in a ratio of 9:8. The ratio of the average experience level of designers to developers is 5:6. What is the average experience level of all the developers in the team (in years)?

(A) 15 (B) 16
(C) 17 (D) 18

Q28 A train stops for 12 minutes each hour during a 5-hour journey. The average speed of the train, including the halts, is recorded as 48 kmph. Determine the average speed of the train excluding the halts (in kmph).

Q29 After covering 660 kilometers from Varanasi, the train experiences a breakdown. It takes 2 hours and 40 minutes to resume operation, thereafter speed gets reduced by of 22.22%. Consequently, it arrives at New Delhi 4 hours behind schedule. If the same issue had arisen after the train had traveled 870 kilometers from Varanasi, it would have been delayed by 3 hours and 20 minutes, including the same time it took earlier to resume operation. What was the originally scheduled duration for the train to reach New Delhi?

(A) 12 hours
(B) 13 hours 20 mins
(C) 15 hours 20 mins
(D) 16 hours

Q30 In a 600-meter race, A beats B by 75 meters. In a 3-kilometer race, A gives B a head start of x meters and still wins by 8 seconds. If A's speed exceeds 108 km/h, what is the maximum possible integer value of x?

Answer Key

Q1	335
Q2	400
Q3	4
Q4	C
Q5	820
Q6	B
Q7	840
Q8	20
Q9	18
Q10	B
Q11	2
Q12	B
Q13	77
Q14	30
Q15	D
Q16	C
Q17	A
Q18	A
Q19	C
Q20	125
Q21	72
Q22	D
Q23	C
Q24	A
Q25	9
Q26	3
Q27	C
Q28	60
Q29	A
Q30	164

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Let A beat B by 'a', here a = 100

B beat C by 'b', here b = 200

A beat C by 'c', here c = 290

the total distance is given by = $\frac{ab}{a+b-c}$

Thus, total distance = $\frac{200\times100}{10}$ = 2000

Thus, the length of the track is $2000m$ In the same racing track P beats Q by $200m$, Q beats R by $150m$ Let, P beats R by $kmetre$

So,

$$k = 200 + 150 - \frac{(200 \times 150)}{2000}$$
$$\Rightarrow k = 350 - 15 = 335$$

Thus, the distance by which P will beat R is $335m$.

Video Solution:

Q2 Text Solution:

Let the track length be t meters

Let the speeds of Virat and Rohit be v m/min and r m/min respectively

Virat takes 2 more laps as compared to Rohit, so

$\frac{40}{\frac{t}{v}} - \frac{40}{\frac{t}{r}} = 2$

$\Rightarrow \frac{v-r}{t} = \frac{1}{20}$

$\Rightarrow t = 20(v-r)$

Also when moving in opposite direction they meet in 4 minutes, so

$\frac{t}{v+r} = 4$

$\Rightarrow t = 4(v+r)$

So, using both the equations, we get

4(v+r) = 20(v-r)

$\Rightarrow \frac{v}{r} = \frac{3}{2}$

So to find the time it takes for Virat to complete a lap, lets find out how many laps he can make in an hour(60 minutes)

$\frac{60}{\frac{t}{v}} = \frac{60}{\frac{4(v+r)}{v}} = 9$

So time required for one lap=

$\frac{60}{9} = 6\ min\ 40\ sec = 400\ seconds$

Video Solution:

Q3 Text Solution:

In a circular track, if two people met at the same point every time they pass each other, then that point has to be their starting point. If they meet at any other point other than the starting point, there will be at least 2 different meeting points.

As they are meeting at the starting point for every time, they should be jogging in the same direction. If they jog in the opposite directions then they should meet each other at a point which is different from the starting point which is not possible.

So, Trisha & Bhavna can only jog in the same direction.

As they meet at the starting point in every 20 s, the relative speed is $\frac{40m}{20s}$ = 2 m/s.

Now let's assume that Trisha's speed is 10x m/s

So, 2x = 2

- x = 1

Hence, Trisha's speed is 10 m/s.

So, she will be able to complete 1 round of the track in $\frac{40\ m}{10\ m/s}$ = 4s.

Video Solution:

Q4 Text Solution:

Let their speeds be T km/hr and P km/hr. (P > T because, as per the data, he crosses T)

Here, we must understand one very important concept of relative motion across circular tracks.

If two people travel in the same direction around a circular track, every time the faster person overtakes the slower person, he covers one round more than the slower one.

So here, when they start running in the same direction, Pumba completes 5 more rounds than Timon in 50 minutes.

OR Pumba would complete 1 more round that timon in 50/5 = 10 mins.

OR When they both travel in the same direction, the time taken to meet is 10 mins or 1/6 hours.

So, **Time x relative speed = distance**

(1/6) hr = 6 KM/P-T

P – T = 36 km/hr -----(1)

Also, when they travel in the opposite directions, Timon covers 1 kilometer less than Pumba.

Let Timon cover a km, Pumba would over 1 km more i.e. a+1.

OR (a) + (a+1) = 6

OR 2a + 1 = 6

OR a = 2.5, a+1 = 3.5 km

In the same time, the ratio of speeds = ratio of their distances

T : P = 2.5 : 3.5

T : P = 5 : 7

T = 5x and P = 7x , put in (1):

7x – 5x = 36

OR 2x = 36

OR x = 18 km.

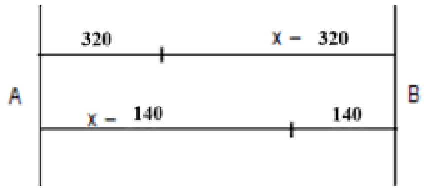
Thus, speeds of T and P are

5 x18 = 90 km/hr

7 x 18 = 126 km/hr

Video Solution:

Q5 Text Solution:



If x is the width of the pond (in metres), then speed at which they travel is proportional to the distances they travel. Hence from the diagram, we get

$$\frac{320}{(x-320)} = \frac{(x-320+140)}{(320+x-140)}$$

x (x - 820) = 0

x = 820

Video Solution:

Q6 Text Solution:

Let the length of the track be t meters and let the speeds be x m/s and y m/s respectively. So

$\frac{t}{x+y} = 25$

$\frac{t}{x-y} = 150$

So

$\frac{x-y}{x+y} = \frac{1}{6}$

$\Rightarrow 5x = 7y$

$\Rightarrow \frac{x}{y} = \frac{7}{5}$

So the ratio of speeds is 7:5

So if they are running in the same direction, they will meet at the starting point for the first time in their second meeting and if they are running in the opposite direction they will meet at the starting point for the first time in their twelfth meeting.

So the first meeting at the starting point will be same in both case, i.e., =

$12 \times 25 \ = \ 2 \times 150 \ = \ 300 \ s \ = \ 5 \ min$

Video Solution:

Q7 Text Solution:

Ratio of speed for P, Q and R = 1200 : 1120 : 1000 = 30:28:25

So ratio of time taken for them to cover equal distance

P:Q:R =

$\frac{1}{30} : \frac{1}{28} : \frac{1}{25} = 700 \ : 750 \ : 840 \ = \ 70 \ : \ 75 \ : \ 84$

Let the time taken by them be 70t, 75t and 84t, so time difference between R and Q is 9t and hence

9t=9 min

⇒t= 1 min

So time difference between R and P = 84t-70t = 14t = 14 min = 840 s

Video Solution:

Q8 Text Solution:

According to the first condition, Rohan runs 1080 meters in 12 seconds more than the time required for Arjun to run 1200m.

So if speeds of Rohan and Arjun are considered to be r m/s and a m/s respectively, then

$\frac{1080}{r} - \frac{1200}{a} = 12$

In the second scenario, Rohan runs 1140 meters in 16 seconds more than the time required for Arjun to run 1200 meters, so

$\frac{1140}{r} - \frac{1200}{a} = 16$

Using both the statements

$$\frac{1140}{r} - \frac{1080}{r} = 16 - 12$$

$$\Rightarrow \frac{60}{r} = 4$$

$$\Rightarrow r = 15\ m/s$$

So

$$\frac{1080}{r} - \frac{1200}{a} = 12$$

$$\Rightarrow 72 - 12 = \frac{1200}{a}$$

$$\Rightarrow a = 20\ m/s$$

Video Solution:

Q9 Text Solution:

Since X and Y are at diametrically opposite points, they would be at a distance of 250 m from each other. So for the first meeting they need to cover 250 m, so time required for first meeting = $\frac{250}{15+10} = 10\ s$

However afterwards they need to cover 500 meters for the meetings afterwards, so the time required = $\frac{500}{15+10} = 20\ s$

So time left after first meeting = 350 - 10 = 340 s

So number of meetings after the first one = $\frac{340}{20} = 17$

So they will meet 17 times after the first meeting meaning they meet 18 times in total.

Video Solution:

Q10 Text Solution:

Ratio of speeds of X, Y and Z = 800 : 760 : 700 = 40: 38 : 35

Now X, Y and Z are having running tracks of 430 m, 400 m and 380 m respectively.

Lets consider the scenario when X reaches the finish line.

If X reaches the finish line, 40 parts = 430 m

So multiplying factor = $\frac{430}{40} = 10.75$

So distance Y will run in the same time =

$38 \times 10.75 = 408.5\ m$

and distance Z will run in the same time =

$35 \times 10.75 = 376.25\ m$

Which means Y will reach the finish line before X and Z and hence will win.

Video Solution:

Q11 Text Solution:

$A : B = 1 : 2$

B : C = 2 : 5

$A : D = 1 : 4$

Thus, $A : B : C : D = 1 : 2 : 5 : 4$

$$\frac{(x + 2x + 5x + 4x)}{4} = \frac{12x}{4} = 3x = 84$$

Therefore, $x = 28$

Marks in subject $A = 28$

Marks in subject $B = 56$

Marks in subject $C = 140$

Marks in subject $D = 112$

Therefore, the student scored equal to or more than 60 marks in only two subjects.

Video Solution:

Q12 Text Solution:

Let the present ages of A and B are $7x$ years and $5x$ years respectively

Let the present ages of C and D are $2y$ years and $3y$ years respectively

According to question:

$$7x + 5x + 3y = 36 \times 3$$
$$12x + 3y = 108$$
$$4x + y = 36$$

Also, $7x + 2y + 3y = 34 \times 3$

$$7x + 5y = 102$$

On solving both the equations, we get $x = 6$ and $y = 12$

Present age of $A = 7 \times 6 = 42$ years

Present age of $D = 3 \times 12 = 36$ years

So, the desired difference $= 42 - 36 = 6$ years

Video Solution:

Q13 Text Solution:

Total weight of A, B and C is

$(65 \times 3) = 195kg$

Total weight of A, B, C and D is

$(68 \times 4) = 272kg$

Weight of $D = (272 - 195) = 77kg$

Weight of $E = 77 - 4 = 73kg$

Total weight of B, C, D and $E = (67 \times 4) = 268kg$

Total weight of B, C and $D = 268 - 73 = 195kg$

Weight of $A = (272 - 195) = 77kg$.

Video Solution:

Q14 Text Solution:

Let total number of students in section 'A' be 'x'

Number of students in section 'B' and 'C' together $= 120 - x$

So,

$48 \times x + (120 - x) \times \frac{170}{3} = 120 \times 54.5$

Or, $48x + \frac{20400}{3} - \frac{170x}{3} = 6540$

Or, $\frac{26x}{3} = 260$

Or, $x = 30$

Alternatively, the above problem can also be approached using Alligation method.

A **B + C**

48 $\frac{170}{3}$

54.5

$\frac{13}{6}$ 6.5

which gives us the ratio as 1 : 3

So, the number of students in section A =

$\frac{1}{4} \times 120 = 30$

Video Solution:

Q15 Text Solution:

Let ages of 'A', 'B', 'C' and 'D' be 'a' years, 'b' years, 'c' years and 'd' years, respectively such that a > b > c > d as they all have distinct ages.

Thus we have,

a + b + c = 43 x 3 = 129 (Equation 1)

a + b + d = 42 x 3 = 126 (Equation 2)

a + c + d = 38 x 3 = 114 (Equation 3)

b + c + d = 33 x 3 = 99 (Equation 4)

(Equation 1) - (Equation 4)
a - d = 30 years

Video Solution:

Q16 Text Solution:

Let the number of students in class X and class Y be a and b respectively.
We know that the average weight of both the classes combined is $75kg$.
Total weight of the 2 classes combined
$= 75(a+b) = 75a + 75b$
The average weight of class X is $45kg$ more than the average weight of class Y.
Let the average weight of class Y be ' x '.
Total weight of the 2 classes combined

$$= (x+45)a + xb$$

Combining (1) and (2), we get,

$$\begin{aligned}
&75a + 75b = (x+45)a + xb \\
&75a + 75b = xa + 45a + xb \\
&75a - 45a - xa = xb - 75b \\
&30a - xa = xb - 75b \\
&a(30-x) = b(x-75) \\
&\frac{a}{b} = \frac{(x-75)}{(30-x)}
\end{aligned}$$

It has been given that x is an integer.
Option A:

$$\begin{aligned}
&\frac{(x-75)}{(30-x)} = \frac{8}{1} \\
&x - 75 = 240 - 8x \\
&9x = 315 \\
&x = 35
\end{aligned}$$

Option B:

$$\begin{aligned}
&\frac{(x-75)}{(30-x)} = \frac{1}{2} \\
&2x - 150 = 30 - x \\
&3x = 180 \\
&x = 60
\end{aligned}$$

Option D:

$$\begin{aligned}
&\frac{(x-75)}{(30-x)} = \frac{2}{1} \\
&x - 75 = 60 - 2x \\
&3x = 135 \\
&x = 45
\end{aligned}$$

Option C:

$$\begin{aligned}
&\frac{(x-75)}{(30-x)} = \frac{1}{5} \\
&5x - 375 = 30 - x \\
&6x = 405 \\
&x = 67.5
\end{aligned}$$

Video Solution:

Q17 Text Solution:

Let the average marks scored by the initial 25 boys be a and the average marks scored by the initial 25 girls be b and the average marks scored by the new students be c

$$25a + 10c = 35 \times 1.2a = 42a$$

$$\Rightarrow 10c = 17a\,.......(1)$$
$$25b + 10c = 35 \times .8b = 28b$$
$$\Rightarrow 10c = 3b\,..........(2)$$

Equating equation (1) and (2), we get

$17a = 3b\ \frac{a}{b} = \frac{3}{17}$ or $a : b = 3 : 17$

Video Solution:

Q18 Text Solution:

Let, the average weight of the remaining 8 students be x kg.

Average weight of the two heaviest students =

$14\frac{6}{11}$% more than (x + 3) kg

= (1+ $\frac{160}{1100}$) × (x+3) kg

= $\frac{63(x+3)}{55}$

= 10(x+3) - 8x = 2× $\frac{63(x+3)}{55}$

= (2x + 30) 55 = 126x+ 378

16x = 1272

x = 79.5

Video Solution:

Q19 Text Solution:

Let the number of vans originally estimated be v

So each van was initially planned to carry =

$\frac{120}{v}$ boxes

Each van actually carried =

$\frac{120}{v} - 2 = \frac{120-2v}{v}$ boxes

So

$$\left(\frac{120-2v}{v}\right)\left(v+5\right) = 120$$
$$\Rightarrow 2v^2 + 10v - 600 = 0$$
$$\Rightarrow v = -20 \text{ or } 15$$

As the number of vans cannot be negative, there were 15 vans planned initially.

Video Solution:

Q20 Text Solution:

Let the numbers be n_1, n_2, n_3,..., n_{19}

So

$n_1+n_2+n_3+...+n_{19}=35\times19 = 665$

Also

$n_1+n_2+n_3+...+n_{10}=21\times10 = 210$

and

$n_{10}+n_{11}+n_{12}+...+n_{19}=48\times10 = 480$

So

$(n_1+n_2+n_3+...+n_{10})+(n_{10}+n_{11}+n_{12}+...+n_{19})=$

$(n_1+n_2+n_3+...+n_{19})+n_{10}$

$\Rightarrow 210+480=665+n_{10}$

$\Rightarrow n_{10}=25$

To maximise eighteenth number we should minimize eleventh to seventeenth, so they will be 26, 27, 28, 29, 30 and 31

So

$n_{10}+n_{11}+n_{12}+...+n_{19}= 480$

$$\Rightarrow 25 + 26 + 27 + 28 + 29 + 30 + 31$$
$$+ 32 + n_{18} + n_{19} = 480$$
$$\Rightarrow n_{18} + n_{19} = 252$$

As we have to maximize n_{18}, difference between n_{18} *and* n_{19} should be minimum. Since difference of 1 will result in them being non natural, hence we will take a difference of 2 and hence the eighteenth number will be 125 and nineteenth will be 127.

Video Solution:

Q21 Text Solution:

Let the initial average height be h inches,so

Initial total height = nh inches

New total height = (n+4)(h+3) = (nh+3n+4h+12) in

So

$nh + 3n + 4h + 12 - nh = (25 \times 12)$

$\Rightarrow 3n + 4h = 288$

$\Rightarrow h = 72 - \frac{3}{4}n$

If n = 4 , h=69 in

If n = 8 , h=66 in

So if n keeps on increasing, the average height will keep on decreasing, so maximum possible average height is 69 inches, and hence the average of the group after 4 players join = 69+3 = 72 inches

Video Solution:

Q22 Text Solution:

Let the number of exams taken by Ravi, Mohan and Suresh be r, m and s. So,

$\frac{67r+81m}{r+m} = 73$

$\Rightarrow 67r + 81m = 73r + 73m$

$\Rightarrow 8m = 6r$

$\Rightarrow \frac{r}{m} = \frac{4}{3}$

$Similarly$

$\frac{81m+70s}{m+s} = 76$

$\Rightarrow 81m + 70s = 76m + 76s$

$\Rightarrow 5m = 6s$

$\Rightarrow \frac{m}{s} = \frac{6}{5}$

So

r:m:s = 8:6:5

So average of all three

$\frac{67r+81m+70s}{r+m+s}$

$= \frac{67\times8+81\times6+70\times5}{8+6+5}$

≈ 72.2

Video Solution:

Q23 Text Solution:

The number of employees in Departments, X : Y : Z = 7 : 8 : 9

Let the average score of Department Z = p

Then, the average score of Department Y = p + 18

Let the average score of Department X = q

Therefore, by the rule of weighted average, we have

$\frac{7q+8(p+18)+9p}{7+8+9} = q - 4$

$\frac{7q+17p+144}{24} = q - 4$

$7q + 17p + 144 = 24q - 96$

$17q - 17p = 240$

$q - p = \frac{240}{17} \approx 14$

So, the approximate absolute difference between the average scores of Departments X and Z is 14.

Video Solution:

Q24 Text Solution:

10 years ago, the total age of 5 members was = 5×40 = 200 years

8 years ago, the total age of (5 members + 2 children) = 200 + 5 × 2 + 2 × 0 = 210 years
(since the children just born in this year.)
7 years ago, the total age of (5 members + 2 children + wife of a member) = 210 + 7 + x = (217 + x) years (say, in this year, the age of the wife of the member was x years.)
5 years ago, the total age of (5 members + 2 children + wife of a member) = 217 + x + 2 × 8 = (233 + x) years
So, according to the condition,

$$\frac{233+x}{8}=32$$

$$233+x=256$$

$$x=23$$

Therefore, 7 years ago, the age of the wife of the newly married member was = 23 years
So, the present age of the wife of the newly married member is = 23 + 7 = 30 years

Video Solution:

Q25 Text Solution:

So the average weight of all the stones =

$$\frac{76+77+79+80+82+83+84+85+86+88}{20}=41\ kg$$

(There are 10 measures with 2 stones each)
So the weight of two heaviest stones = 88
and the weight of the two lightest stones = 76
So the weight of the middle stone =
$(5\times41)-88-76$ = 41kg
The second heaviest weight is going to be the highest weight plus the third highest weight
So letting the weight of the heaviest stone to be x
86 = x+41
⇒x = 45 kg
Similarly the second lightest weight is going to be the lightest weight plus the third lightest weight, so taking the lightest weight to be y
77 = y+41
⇒y = 36 kg
So the difference between the two = 45-36 = 9 kg

Video Solution:

Q26 Text Solution:

$$\mathrm{x_1+x_2+x_3+...+x_n=20n}$$

Also

$$\mathrm{(x_1+20)+x_2+x_3+...+(x_n-5)}$$
$$\mathrm{=25n}$$
$$\mathrm{\Rightarrow x_1+x_2+x_3+...+x_n+15=25n}$$
$$\mathrm{\Rightarrow 20n+15=25n}$$
$$\mathrm{\Rightarrow n=3}$$

Video Solution:

Q27 Text Solution:

The ratio of developers to designers in the team is 9:8.
So, in a team of 68 members, the number of developers is 36 and the number of designers is 32.
The ratio of the average experience of designers to developers is 5:6.
Let the average experience of developers and designers in the team be 6x and 5x respectively.

$$\frac{6x\times36+5x\times32}{68}=188$$

x = 34
The average experience of developers =

$6x = 6 \times 34 = 204$ months $= \frac{204}{12} = 17$ years

Video Solution:

Q28 Text Solution:

Since the train is travelling only for 48 minutes in an hour, so excluding stoppage time the train travels 48 km in 48 minutes, so

Average speed of train excluding the halts =

$\frac{48}{48} \times 60 \;=\; 60\ kmph$

(No use of the time period since the stoppage is constant across the time period)

Video Solution:

Q29 Text Solution:

If the initial speed is assumed to be x km/h,

the new speed will be $\frac{7}{9}x$ km/h.

If the breakdown had occurred after 210 kms, it would have taken 40 minutes more

Therefore,

$\frac{210}{x} = \frac{210}{\frac{7}{9}x} - \frac{40}{60}$

which gives us, x = 90 km/h

As the train had taken 1 hour 20 minutes more to cover the required distance (2 hour 40 minutes was breakdown time), time taken would have been $\frac{9}{7}$ times the originally supposed time.

Hence if time the train was supposed to take to travel the remaining distance is t, then

$\frac{2}{7}$t = 80

t = 280 minutes = 4 hours 40 mins

Time taken to travel 660 kilometers = 660/90 = 7 hours 20 mins

Total time = 12 hours

Video Solution:

Q30 Text Solution:

Let the speeds of A and B be a m/s and b m/s, so

$\mathrm{a} > 108\ \mathrm{km/h}$

$\Rightarrow \mathrm{a} > \frac{5}{18} \times 108$

$\Rightarrow \mathrm{a} > 30\ \mathrm{m/s}$

$\mathrm{In\ 600\ meter\ \ race}$

$\frac{\mathrm{a}}{\mathrm{b}} = \frac{600}{525}$

$\Rightarrow \frac{\mathrm{a}}{\mathrm{b}} = \frac{8}{7}$

$\Rightarrow \mathrm{b} = \frac{7}{8}\mathrm{a}$

Distance ran by b in the second race = (3000-x) m

and time requied by b to cover this distance =

$\left(\frac{3000}{a} + 8\right)\ s$

So

$(3000 - x) = \frac{7}{8}a \times \left(\frac{3000}{a} + 8\right)$

$\Rightarrow 3000 - x = \frac{7}{8} \times (3000 + 8a)$

$\Rightarrow 3000 - x = 2625 + 7a$

$\Rightarrow 375 - x = 7a$

$\Rightarrow 375 - x > 210$

$\Rightarrow x < 165$

So maximum possible integer value of x is 164 m.

Video Solution: